Intelligent Healthcare Service Application

Your selected application may address any domain. It must be designed as a One-Person AI Company with at least three distinct AI roles performing primary stakeholder or business functions. Every group member must individually implement at least two core features and one optional feature. Here is an example of a healthcare application designed to offer patients General Practitioner (GP) consultation services.

In daily life, many people experience minor physical discomfort but hesitate to seek medical attention due to the cumbersome process of traditional consultations. This often results in untreated minor symptoms, which can escalate into more significant health issues over time. To address this gap, we have developed an intelligent healthcare application powered by LLM-based agents and supervised by one human founder/operator. This application aims to provide quick, accurate, and personalized medical advice, minimizing the need for in-person consultations for common ailments. By efficiently assessing and responding to minor health concerns, the application not only enhances patient outcomes but also reduces the strain on healthcare resources.

Primary Users and Roles

The primary users and roles of this application include patients, the AI Care Coordinator, the AI Doctor Agent, the Human Doctor, the AI Technical Manager, and the human founder/operator:

- **Patients** interact with LLM-based agents to describe their symptoms and receive initial health recommendations. If a potential issue is detected, the application can automatically recommend suitable doctors and even schedule appointments based on patient preferences.
- **The AI Care Coordinator** manages patient-facing conversations, symptom intake, appointment coordination, and follow-up services.
- **The AI Doctor Agent** produces and reviews detailed health reports, helping patients and external healthcare professionals understand the case more effectively.
- **The Human Doctor** reviews escalated or low-confidence cases, provides professional diagnoses and treatment, and makes clinical decisions that must not be delegated to AI agents.
- **The AI Technical Manager** manages patient records, permissions, system monitoring, and controlled updates to the LLM-based agents.
- **The Human Founder/Operator** supervises the AI roles, reviews exceptions, and remains responsible for the operation of the company.

This collaborative interaction between patients, the human founder/operator, and the three AI roles ensures that the application delivers comprehensive and efficient healthcare services.

Key Capabilities and Core Features

<u>For Patients</u>

The application is designed to provide efficient and reliable healthcare services through a range of automated features, including diagnosis, health records management, medical appointments, and health consultations. These services are primarily delivered by the AI Care Coordinator to provide a seamless and personalized user experience.

- **Automatic Diagnosis Feature:**
    - Patients engage in interactive sessions with LLM-based agents to describe their symptoms, which the agents use to provide an initial diagnosis.

The system allows patients to refine their symptom descriptions, enabling the agents to offer more tailored diagnostic suggestions.

- Each diagnosis comes with a confidence level, giving patients insight into the reliability of the advice provided.
- This dynamic interaction fosters an ongoing, personalized consultation process that adapts to the patient's needs.

- **Health Record Feature:**
  - Patients can create an account to activate the health record feature, which automatically logs all medical interactions.
  - These records are continuously updated after each consultation, ensuring that patients have a comprehensive and up-to-date medical history at their fingertips.
  - Patients can access and enhance their health records as needed, making it easier to track their health over time and share relevant information with healthcare providers.
- **Medical Appointment Feature:**
  - The system allows patients to input their preferences and availability, which the agents use to match them with suitable doctors for in-person appointments.
  - Appointments can be seamlessly integrated into personal calendars, and patients have the flexibility to modify or cancel them as necessary, ensuring convenience and ease of use.
- **Health Consultation Feature:**
  - Beyond automated diagnosis, patients can seek broader health-related advice by interacting with the LLM-based agents.
  - For more specialized needs, the system facilitates direct communication with doctors on the platform, ensuring that patients receive comprehensive support and tailored support for their healthcare concerns.

For the AI Care Coordinator

The AI Care Coordinator manages hand-offs between the AI Doctor Agent and the Human Doctor, ensuring that cases are routed to the appropriate role based on confidence, complexity, and clinical risk.

- **Doctor Coordination:**
  - The AI Care Coordinator forwards routine cases and diagnostic information to the AI Doctor Agent for automated review.
  - It escalates low-confidence, complex, or high-risk cases to the Human Doctor and coordinates the return of clinical decisions to the patient.

For the AI Doctor Agent

The AI Doctor Agent plays a critical role in this agent-based system by reviewing diagnostic information, preparing reports, and supporting direct consultation between patients and external healthcare professionals.

- **Diagnosis Report Review:**
  - The system categorizes diagnostic reports based on medical specialties, allowing the AI Doctor Agent to review the most relevant information.
  - The AI Doctor Agent evaluates the confidence levels of these reports and provides feedback to the founder/operator, contributing to the continuous improvement of the system.

Additionally, the AI Doctor Agent can flag patients whose reports have lower confidence levels for follow-up by an external healthcare professional, ensuring they receive the necessary care.

- **Medical Records Access:**
  - The AI Doctor Agent has access to categorized diagnostic reports and authorized medical records, providing the comprehensive information needed to support patient care.
  - It can also report discrepancies and request human review, ensuring the accuracy and reliability of the data managed by the agent-based system.

For the Human Doctor

The Human Doctor provides professional clinical oversight for cases that require human judgment and intervention.

- **Escalated Case Review:**
  - The Human Doctor reviews low-confidence, complex, or high-risk cases escalated by the AI Care Coordinator.
  - The Human Doctor confirms or corrects the AI-generated assessment and provides the diagnosis, treatment, or referral decision when required.

For the AI Technical Manager

The AI Technical Manager performs the operational responsibilities previously handled by a technical team, helping the one-person company maintain and continuously improve the agent-based system. Its responsibilities include managing permissions, coordinating controlled updates to the LLM-based agents, and ensuring smooth system operations.

- **Daily Maintenance:**
  - The agent utilizes specialized tools to review system logs and issue announcements, maintaining the integrity and performance of the agent-based system.
  - It is responsible for managing non-private information, ensuring that system maintenance does not compromise user privacy.
- **Permission Management:**
  - The AI Technical Manager can enable or disable features for different user groups, customizing the functionality of the LLM-based agents to meet diverse user needs.
  - It also manages user permissions, ensuring that access levels are appropriately assigned and maintained to secure the system.
- **Model Updates:**
  - The agent can access diagnostic records with low confidence levels and prepare controlled model-update recommendations for approval by the founder/operator.
  - It coordinates specific versions of the LLM-based agents tailored to different functionalities and user groups, ensuring that the system remains up to date.

Optional Features

For Patients

- AI-generated avatars that resemble the patient.

- Video consultation capabilities with doctors.
- Management of payment information, including bills, payment methods, account details, payment history, and electronic invoices.
- Reminder services for specific health-related events.

<u>For the AI Doctor Agent</u>

- Note-taking for specific patients to support follow-up and human review.
- Drafting health-related articles for approval before public release.

<u>For the AI Technical Manager</u>

- Access to necessary operational logs and their corresponding fields.
- Features to facilitate communication and hand-offs among the AI roles.

Please Note:

- Your group may select an application from any domain. The healthcare application above is only an example and is not a required project topic.
- Your application must implement a One-Person AI Company with one human founder/operator and at least three clearly differentiated AI roles that perform stakeholder or business functions and collaborate in the application workflow.
- Every group member must individually implement at least two core features and one optional feature, and the submission must clearly identify each member's contribution.
- Your project should be new and novel, or present a meaningful improvement to an existing application. The features should leverage AI models and capitalize on their strengths; consider building collaborating LLM-based agents rather than merely using a large model directly.
- You are required to present the primary users, AI roles, key capabilities, core features, and optional features of your application using the above example as a guide. Use the Lab Project Description document as a guide for requirement documentation, architecture design, system modeling, and prototype implementation.